

Calculus Homework – Monotonicity and convexity.

Exercise. *Using the tools of differentiation, investigate f for monotonicity and convexity!*

$$(1) f(x) := x^2 + x - 6$$

$$(2) f(x) := 2x^2 - 3x - 4$$

$$(3) f(x) := -3x^2 + 2x - 6$$

$$(4) f(x) := 5x^3 - 4x^4$$

$$(5) f(x) := x^3 - 5x^2$$

$$(6) f(x) := (x + 2)(x^2 - 8x + 16)$$

$$(7) f(x) := 2x^3 - 6x^2 - 2x + 6$$

$$(8) f(x) := x + \frac{1}{x}$$

$$(9) f(x) := x^2 + \frac{1}{x^2}$$

$$(10) f(x) := \frac{5x}{(x + 2)^2}$$

$$(11) f(x) := \frac{3x - 5}{x - 2}$$

$$(12) f(x) := \frac{1}{\sqrt{1 - x^2}}$$

$$(13) f(x) := \frac{1}{\sqrt{x^2 - 1}}$$

$$^*(14) f(x) := \sqrt{\cos(x)}$$

$$^*(15) f(x) := \sin^2(x)$$

$$^*(16) f(x) := 1 - \cos^2(x)$$